- **ENABLING - ORACLE DIGITAL ASSISTANT**

Overview Oracle Digital Assistant:

Oracle Fusion Cloud Applications provides its own digital assistant (FADigitalAssistant) to enhance employee productivity and increase engagement by leveraging conversations to complete tasks and answer queries.

The AI-powered conversational UI lets users communicate with Fusion Applications using their natural language with almost zero learning effort. You can configure the digital assistant and use it.

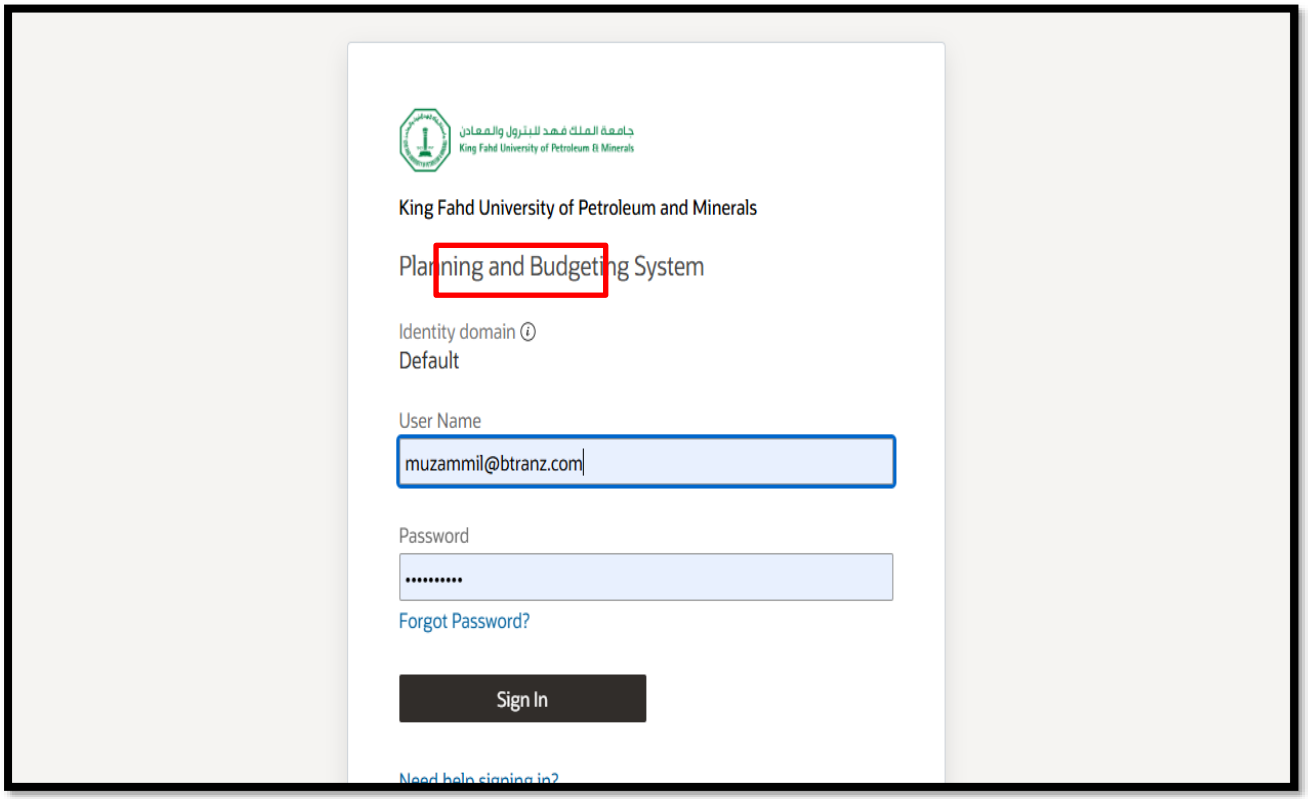
You must train your digital assistant, configure channels, and associate them with your digital assistant. Once you set up your digital assistant, you can test it to check whether it's working correctly on your desktop and mobile devices.

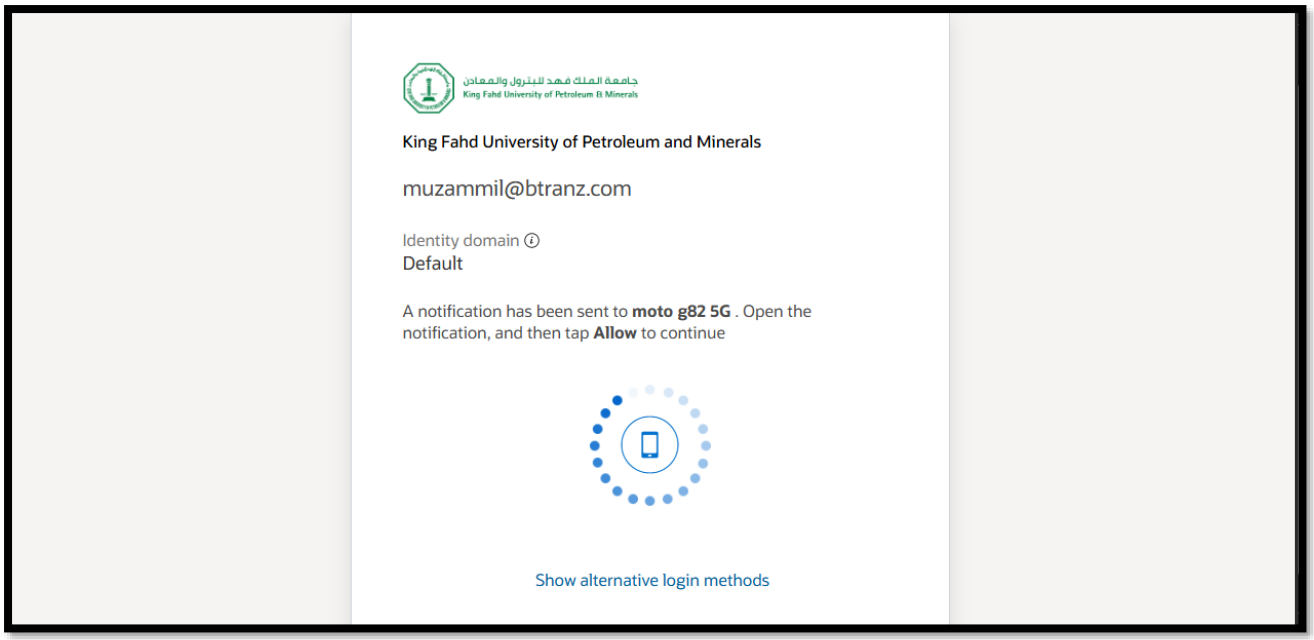
The following steps for Digital Assistant setup:

1. Assign OCI Roles and Responsibilities to user.
2. Digital Assistant User Roles.
3. Configuring FADigital Assistant.
4. Enable Digital Assistant using HCM Design Studio.
5. Roles and Privileges to access Digital Assistant.

1. Assign OCI Roles and Responsibilities to user :

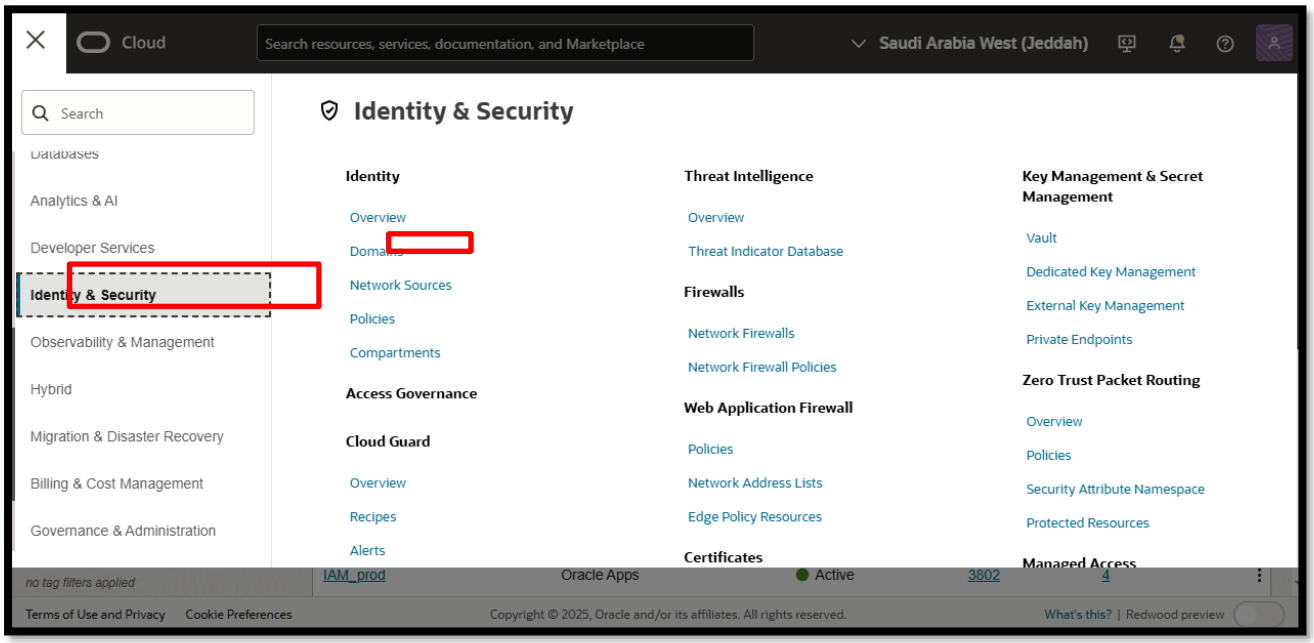
- Login as **OCI (Oracle Cloud Infrastructure)** with OCI Credentials.
- Select Domain as 'Default'.
- A security notification will be sent to allow accessing the OCI console.



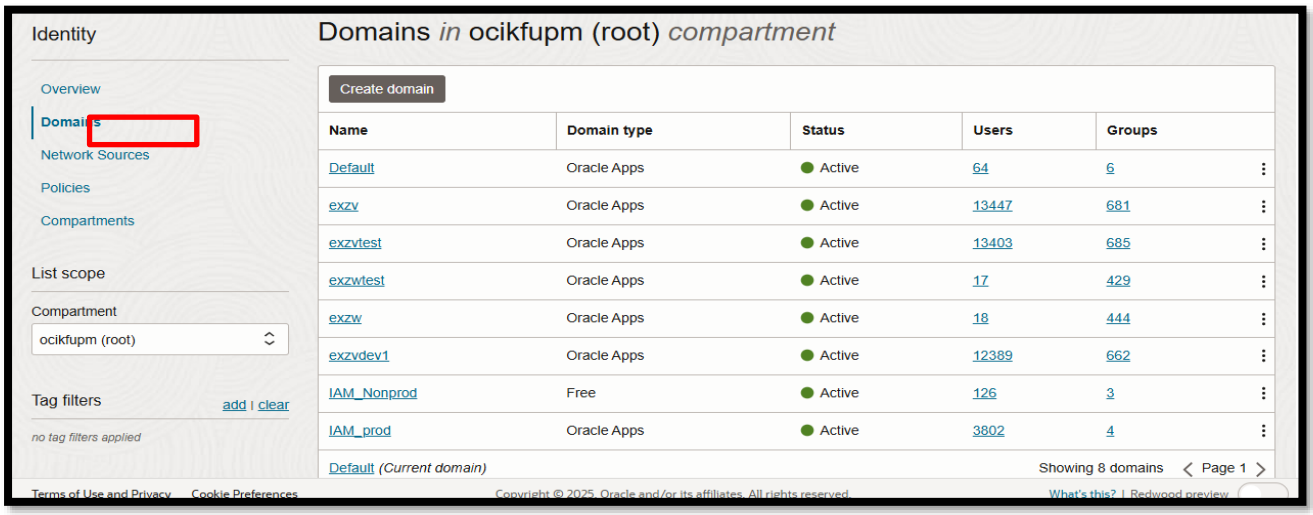


- Under Menu search for 'Identity & Security' and Select 'Domain'

Navigation : MENU→IDENTITY & SECURITY→DOMAIN

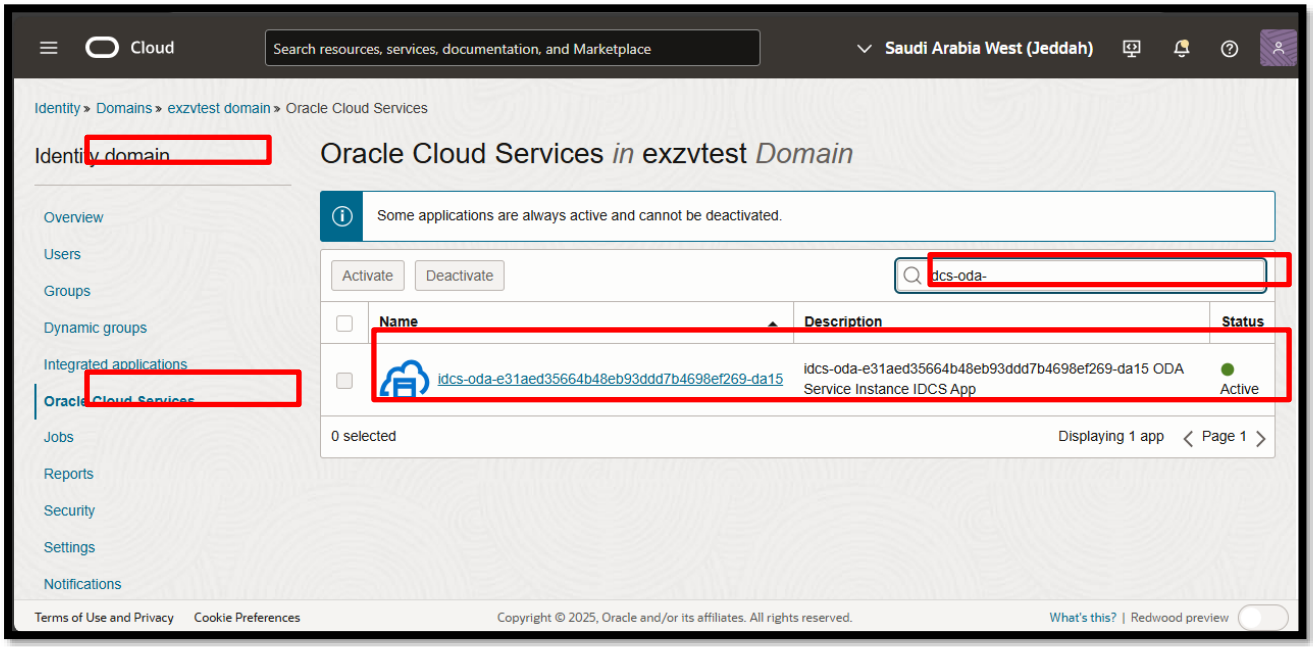


- In the Domains page, select the domain for which you are configuring the Digital Assistant.
- for example if your doing this in Test environment select – **exzvtest**

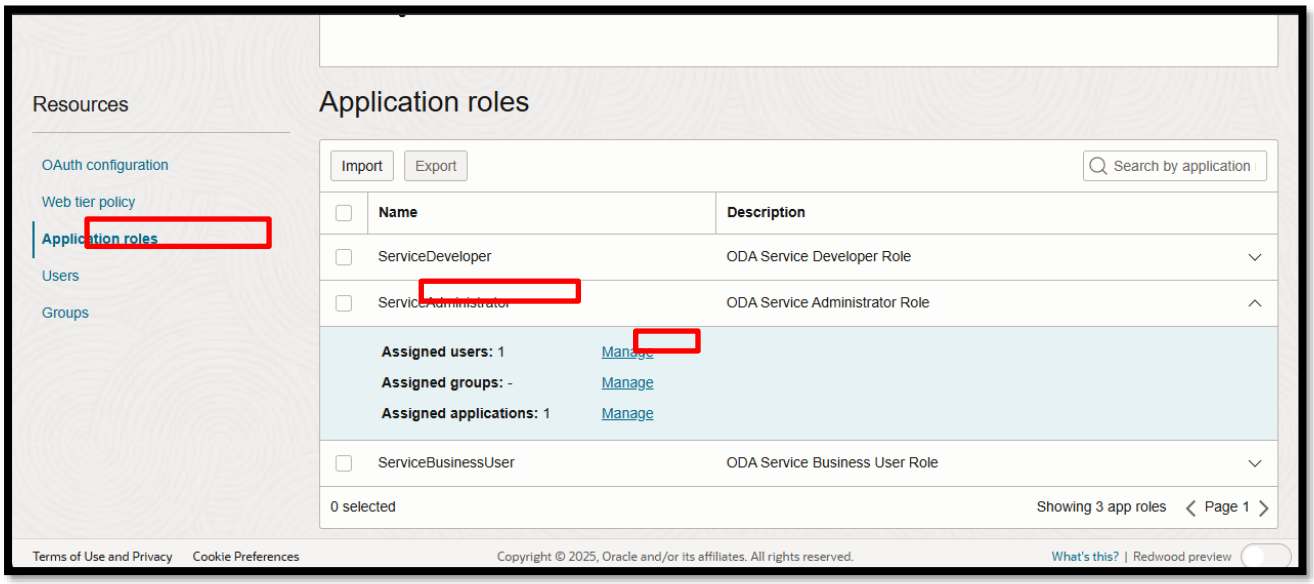


- After selecting your domain, a new page will open when you click on it.under Identity Domain select Oralce Cloud Services, in the search bar on the right side search – **'IDC-ODA'** and click on the IDCS app.

Navigation: Identity Domain -> Oracle cloud services->IDC-ODA

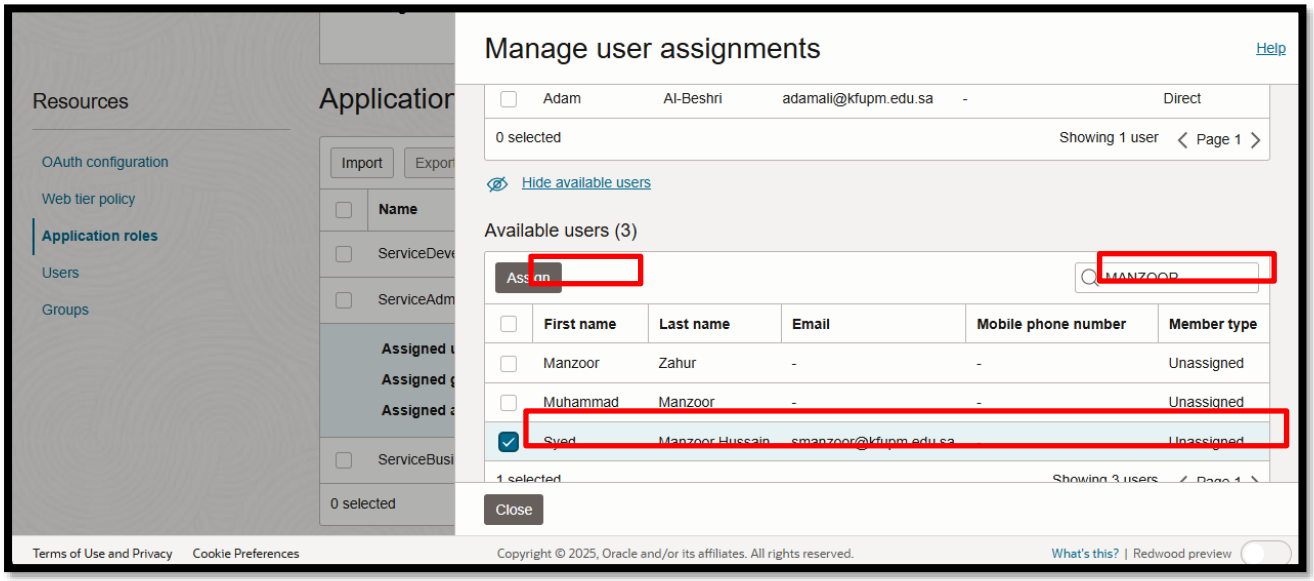


- Under IDCS application, select application roles under Resources to assign user to the "ServiceAdministrator".
- Select Service Administrator and click on Manage…

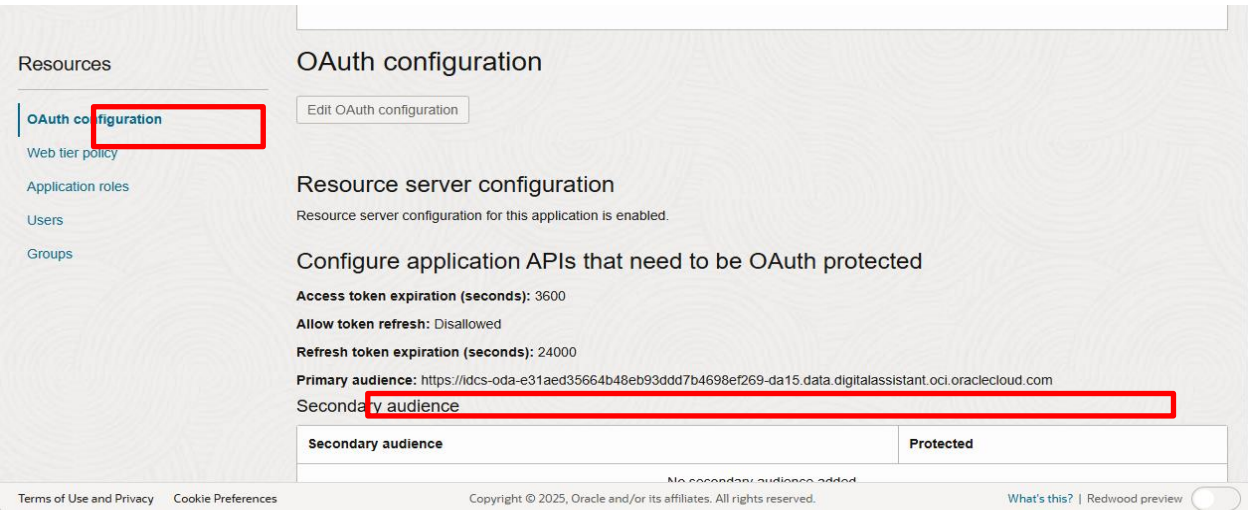


- You can see the users who has already assigned to the role. Click on "Show available users", Search for the username for which you want to give access.

- Select the user name and click on "Assign".



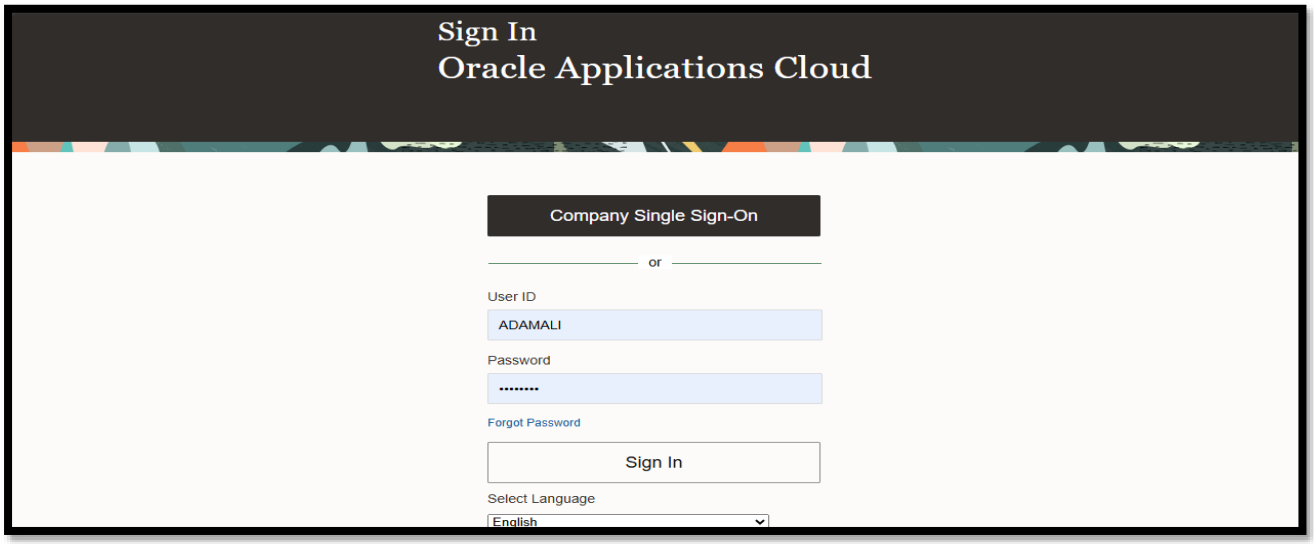
- Back to the application page under resource click on "**OAuth Configuration**", Copy the URL. Using this we can login and configure the digital assistant.



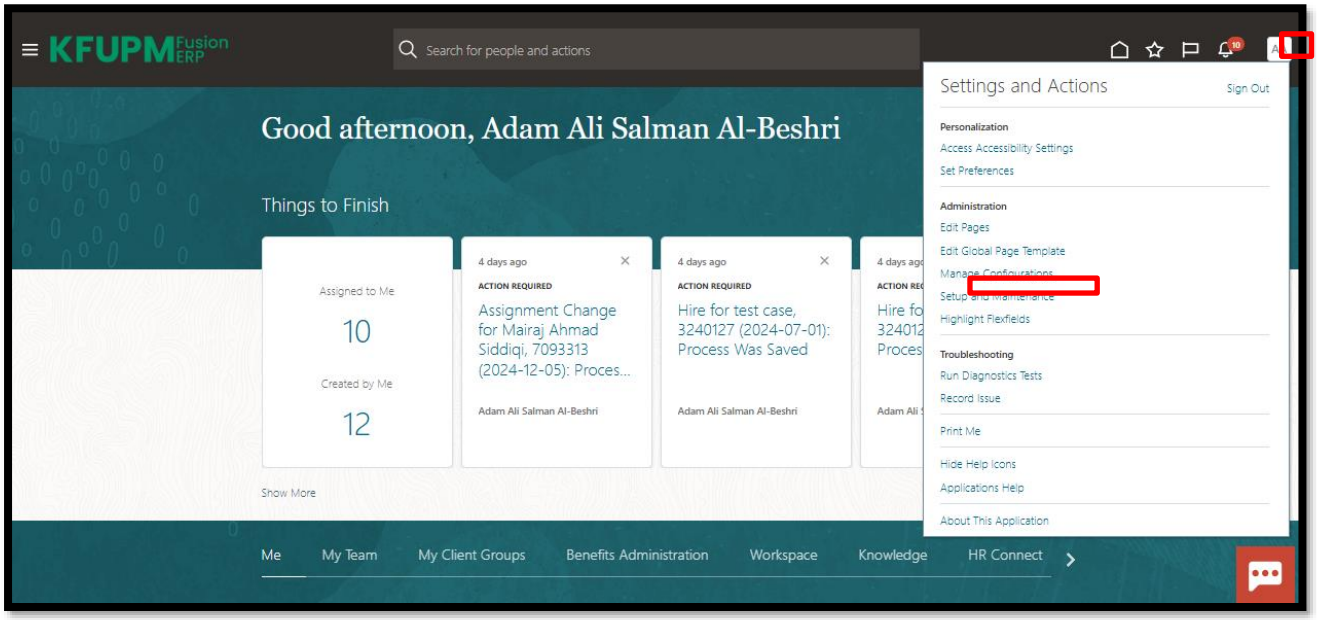
- We have assigned the user to access the IDCS URL. Now, they can either log in through the SaaS platform or directly use the copied link.

2. Steps for Digital Assistant Setup :

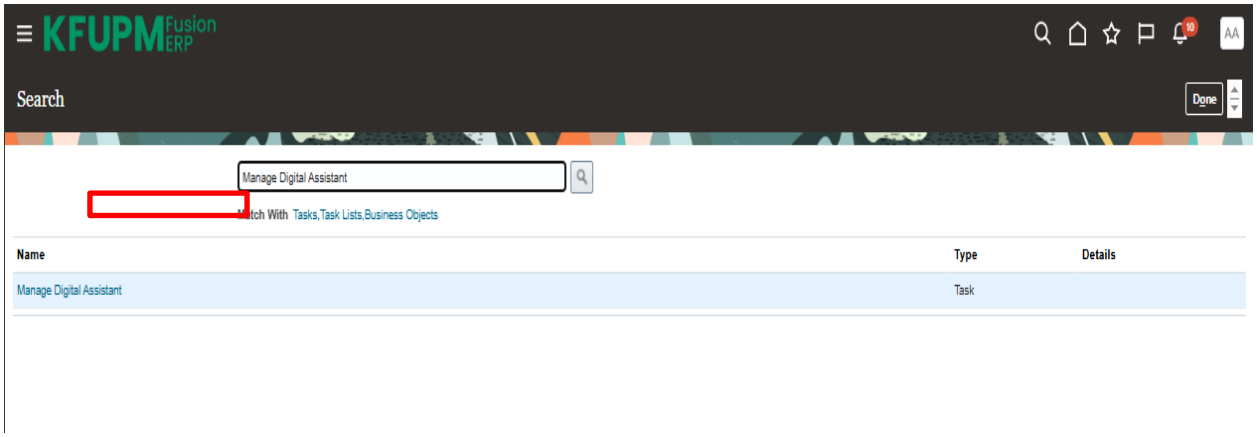
- Login through Saas Platform with the user we assigned in the last step.



- Now, click on FSM, navigate to Setup and Maintenance, and then use the search bar to find "Manage Digital Assistant.

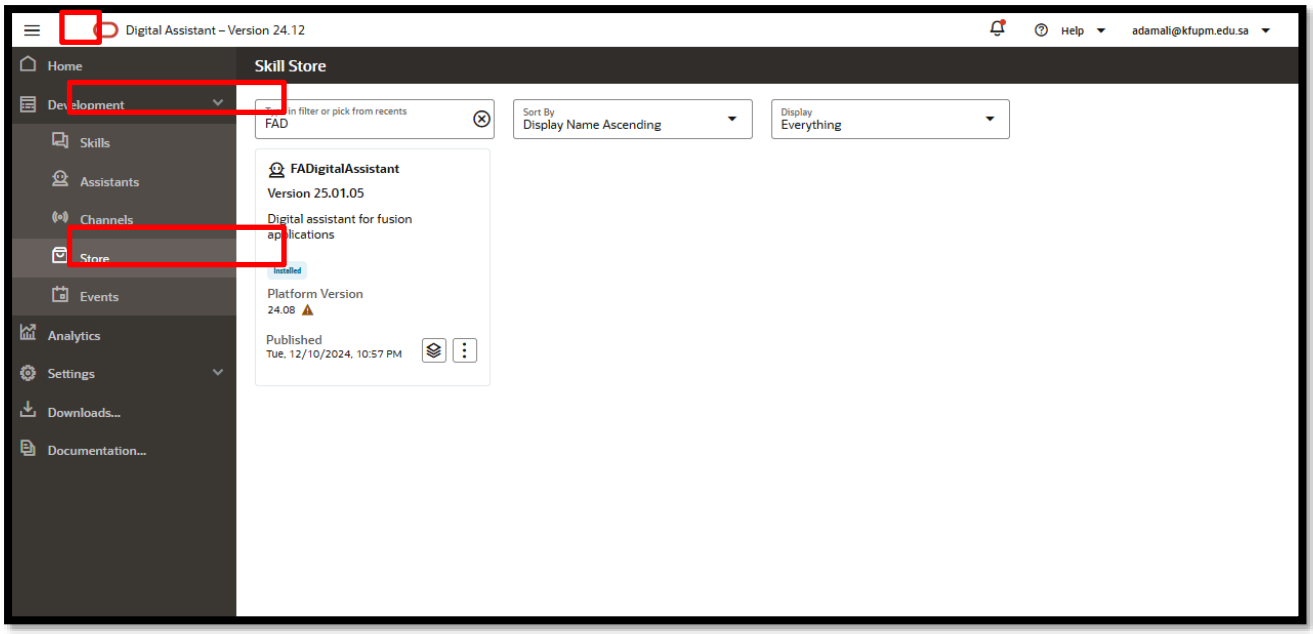


Navigation: FSM→Set & Maintenance→Global Search→ Manage Digital Assistant.

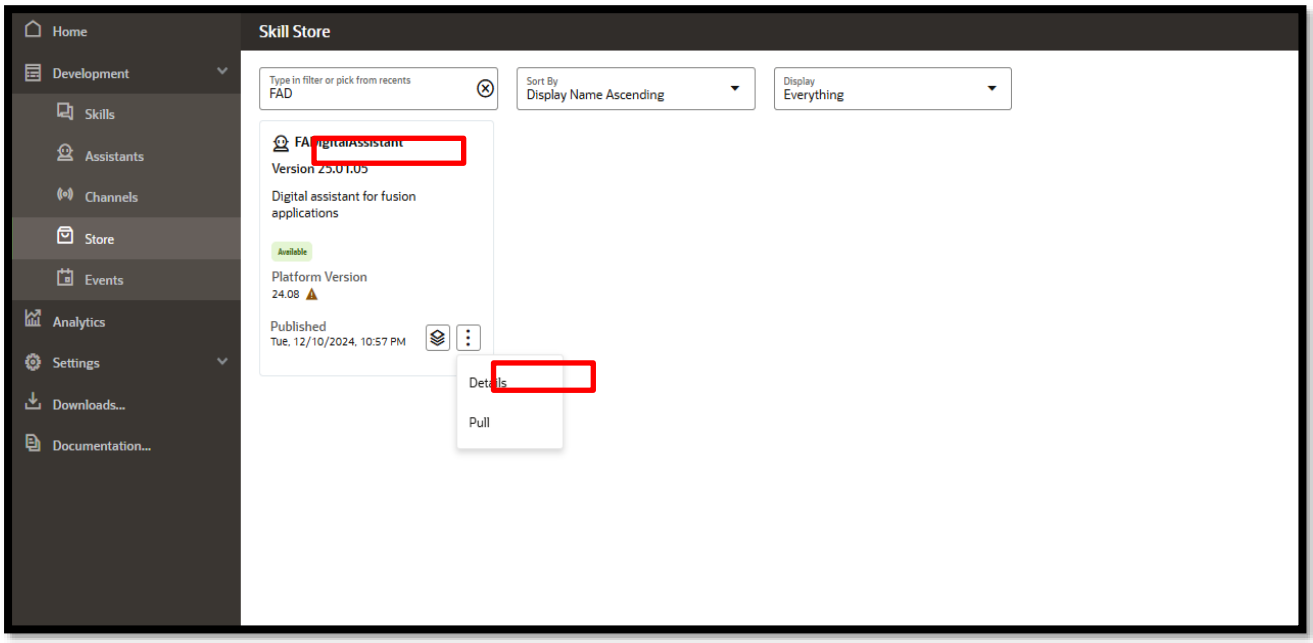


- The new IDCS-ODA page will be opened. This is one way to access the IDCS page or we can directly open the URL which we saved in the step-1.
- You are on the IDCS page logged in as the user assigned to the **serviceAdministrator** role. Now, click on the menu.

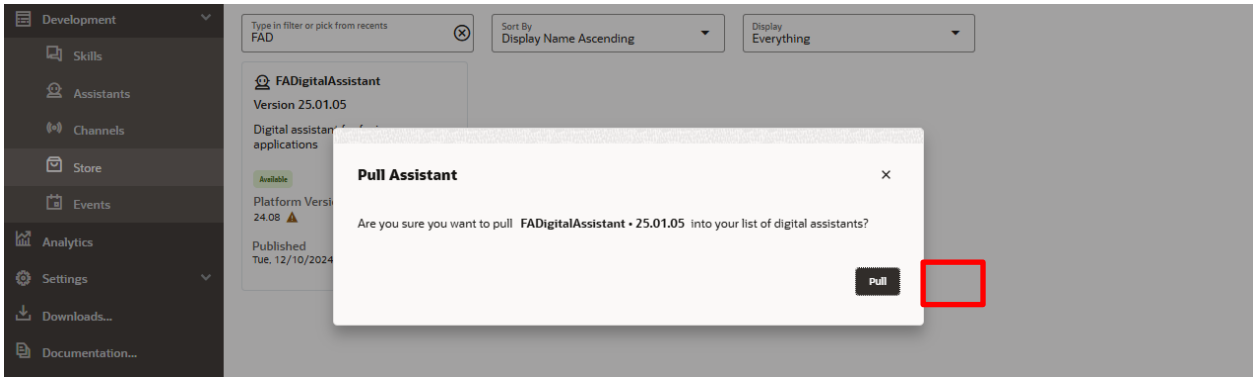
Navigation: Menu→Development→Store→ FADigital Assistant.



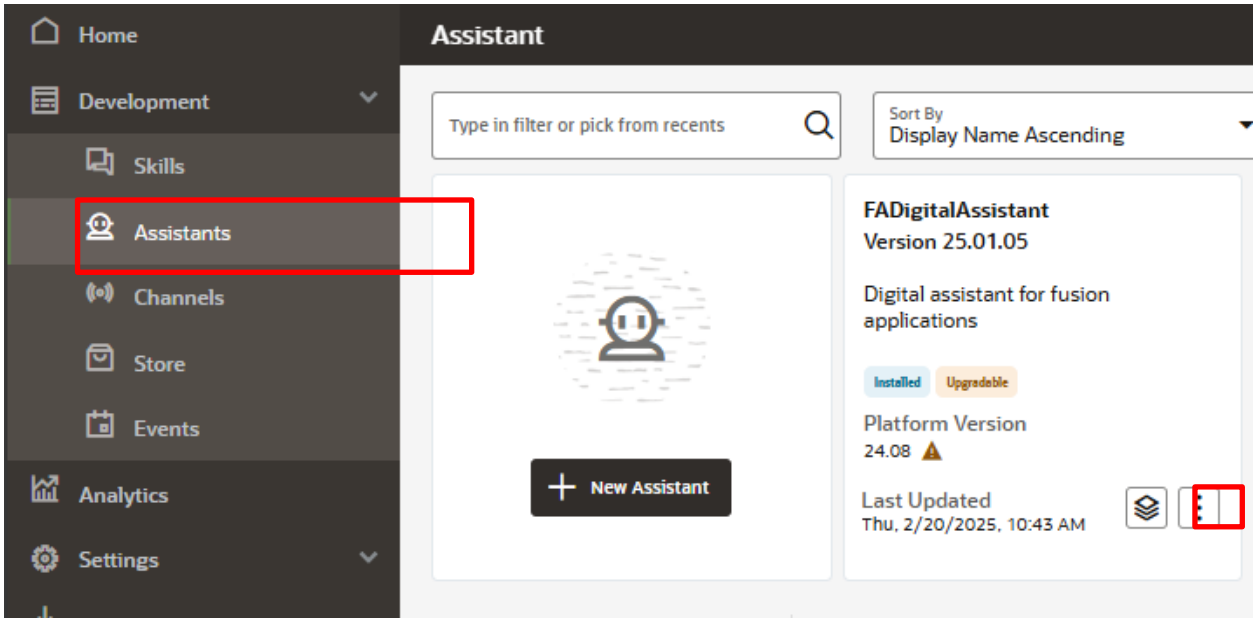
- Now on the Digital assistant, click on three dots click on "**Pull**" to pull your digital assistant. Pull the latest version that is 25.01.05.

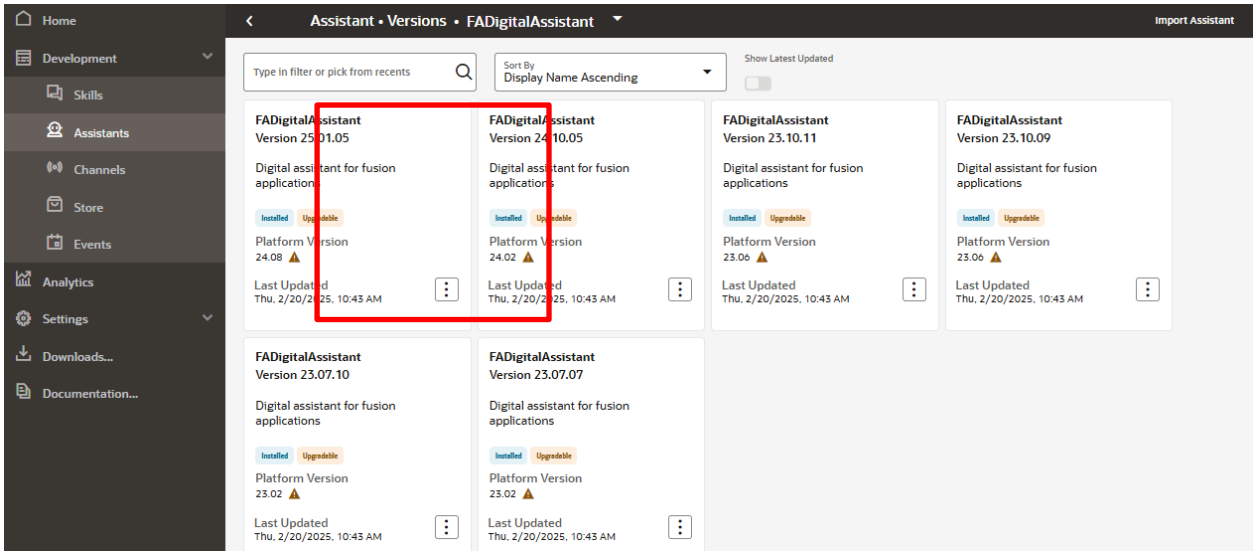


- Click on "**Pull**", latest version of digital assistant will be installed.

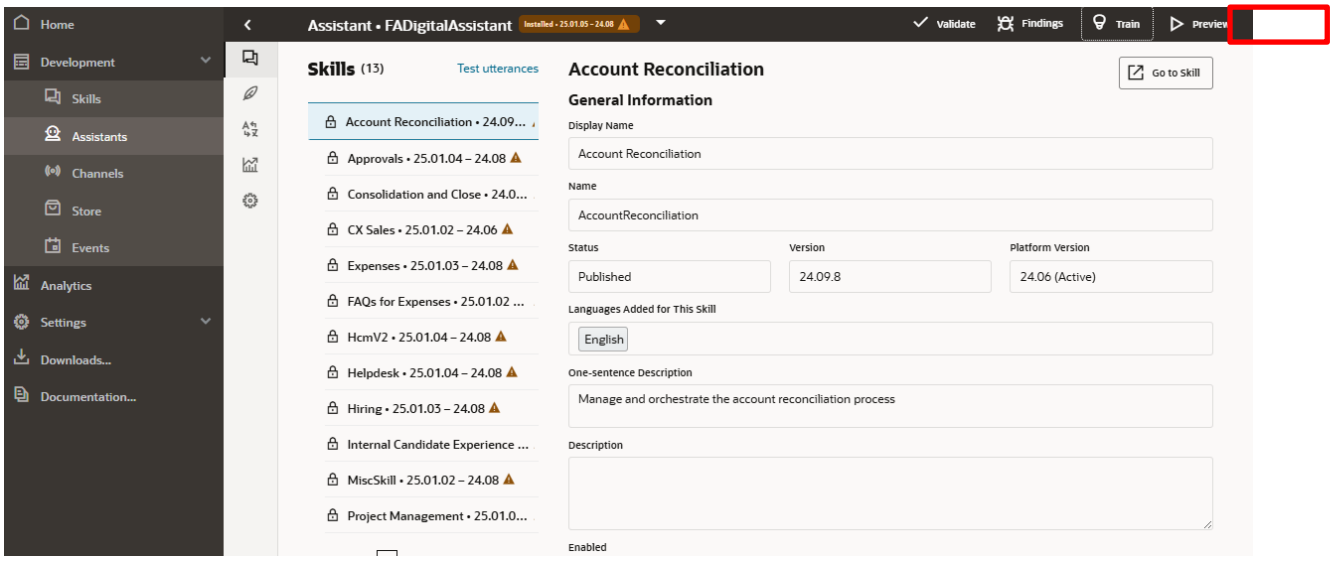


- After completing the installation, navigate to the "Development" section and select "Assistants," where you can view the installed digital assistant.
- Open the latest Digital Assistant, if your unable to see the version 25.01.05, click on the version tab to see all the versions , from there we can select latest version of Digital Assistant.

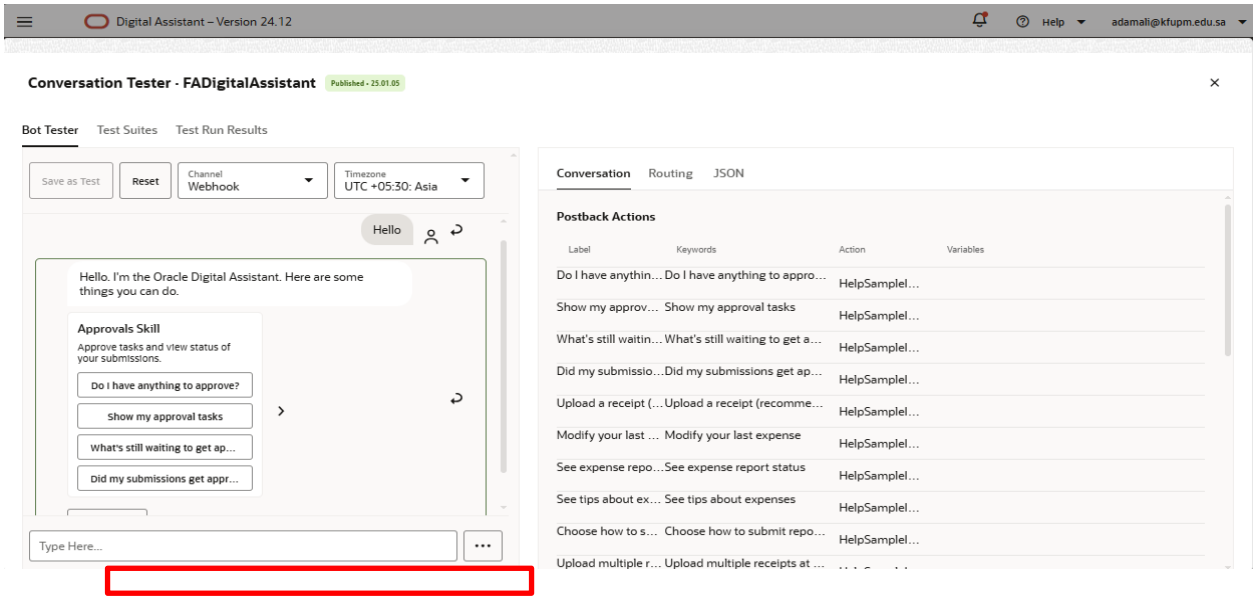




- Open the latest version of Digital Assistant , on the left side you can view the all the skills that are in digital assistant we cannot edit or delete the skills.
- The digital assistant is already trained, and to preview how it works, click on "Preview."



- On the preview page we test Digital Assistant how it is performing. Type Hello..

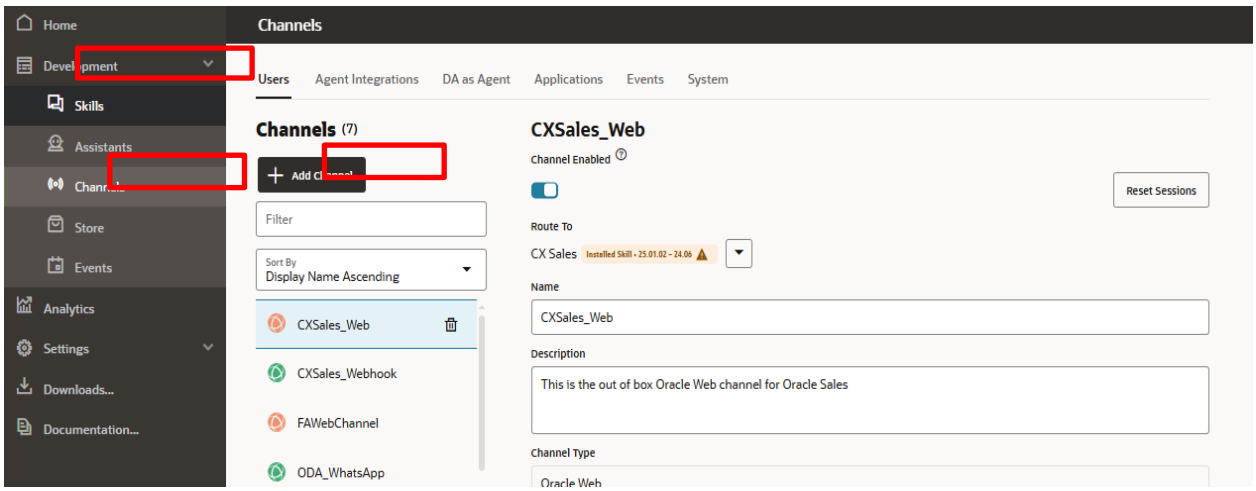


- Digital Assistant is trained and we have tested how its working ,now it's time to create channel for Digital assistant.

3. Steps to create channel :

- You can use your digital assistant in various channels, such as Oracle Web, Twilio, and Microsoft Teams. You can configure these channels to associate them with your digital assistant.
- To configure channel follow the bellow steps..

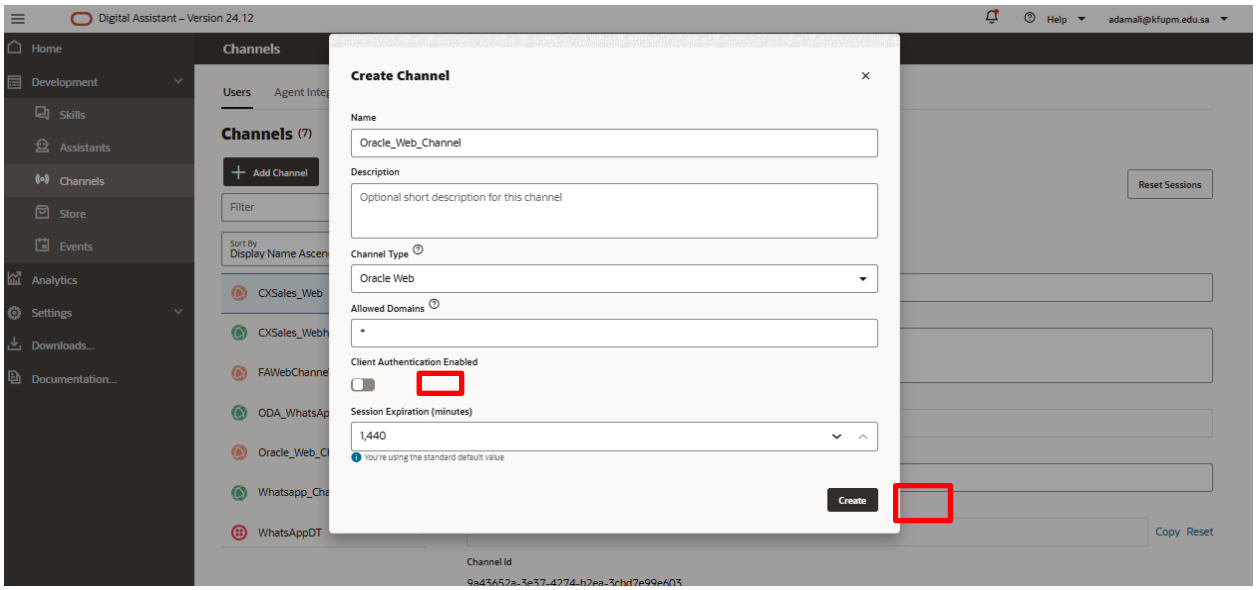
Navigation→Menu→ Development→Channels→Add Channel



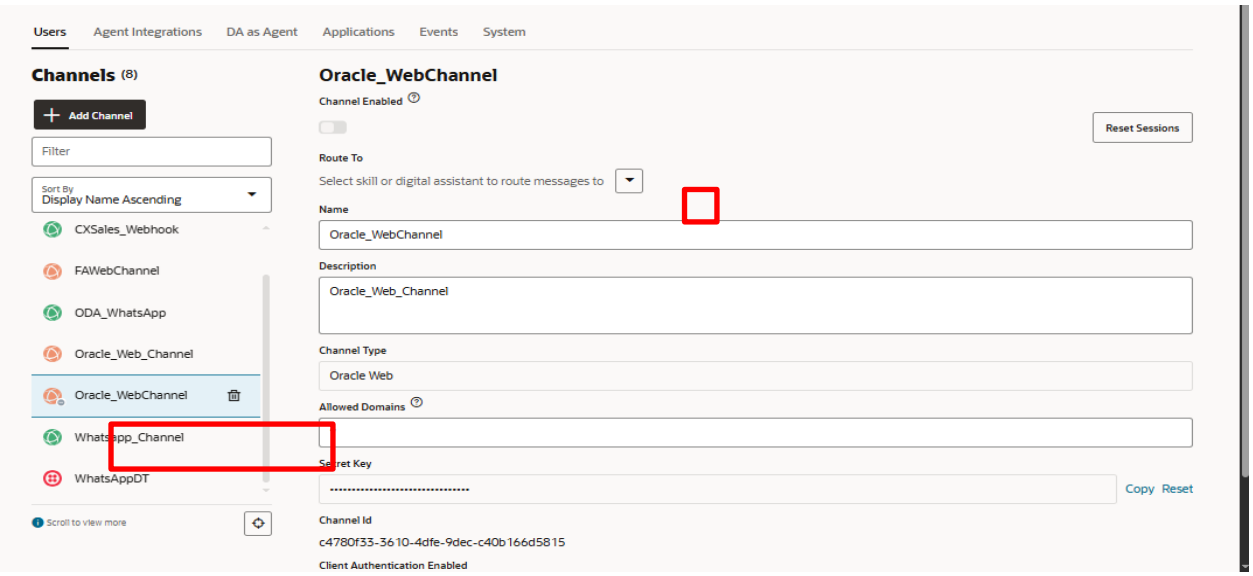
- In the Create Channel dialog box, do these tasks:

a. Enter the name and description of your channel.

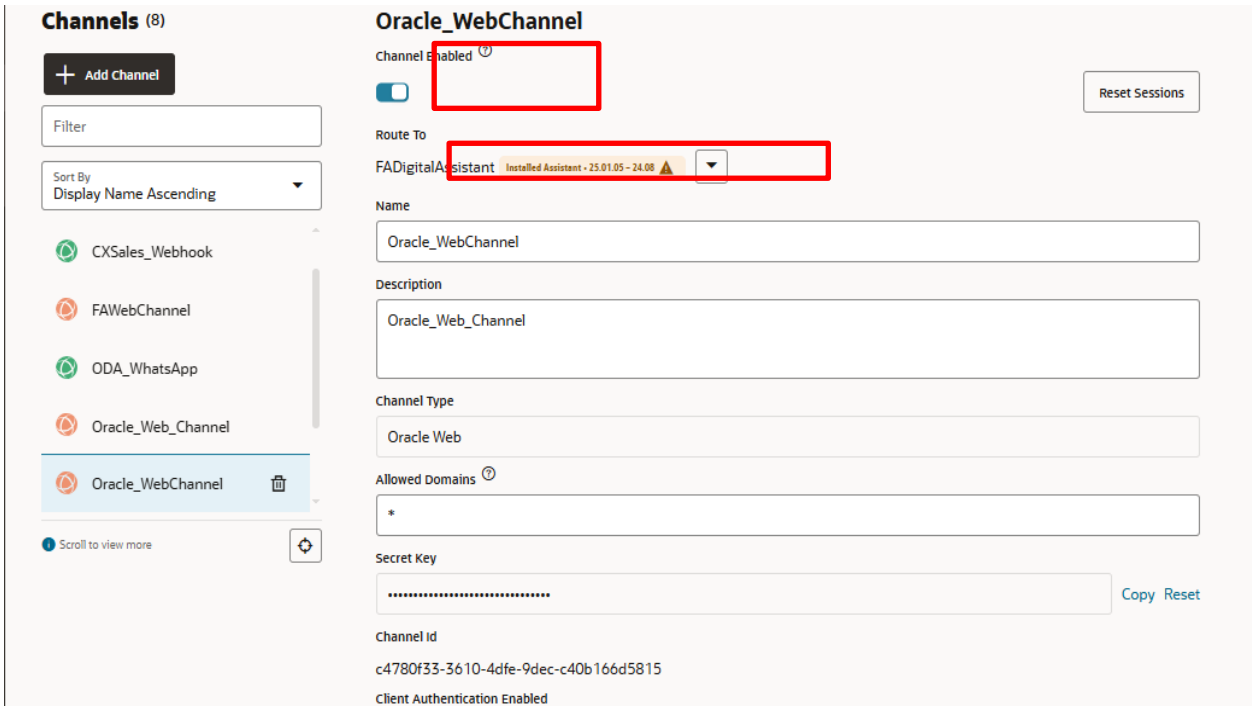
b. From the Channel Type list, select **Oracle Web**.
c. In the Allowed Domains field, enter * or the URL of your application.
d. Turn off Client Authentication Enabled option for unrestricted access.
e. Optionally, set the availability duration for the Oracle Web channel. The default value is 1440 minutes.
f. Click **Create**.



- Now we have to route channel with digital assistant for that click on "**route to**".



- Search for FADigital Assistant Version 25.01.05, Select Digital Assistant.**Enbale the Channel, Click on "Reset Sessions".**



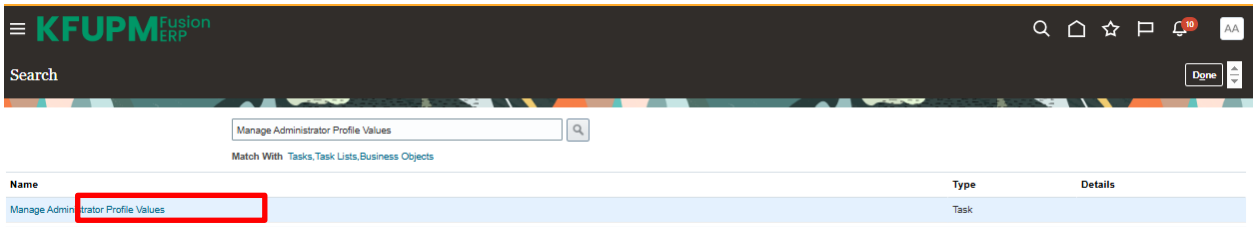
- Copy the channel ID and save it, as we will use it in the next steps.

4. Steps to enable Digital Assistant

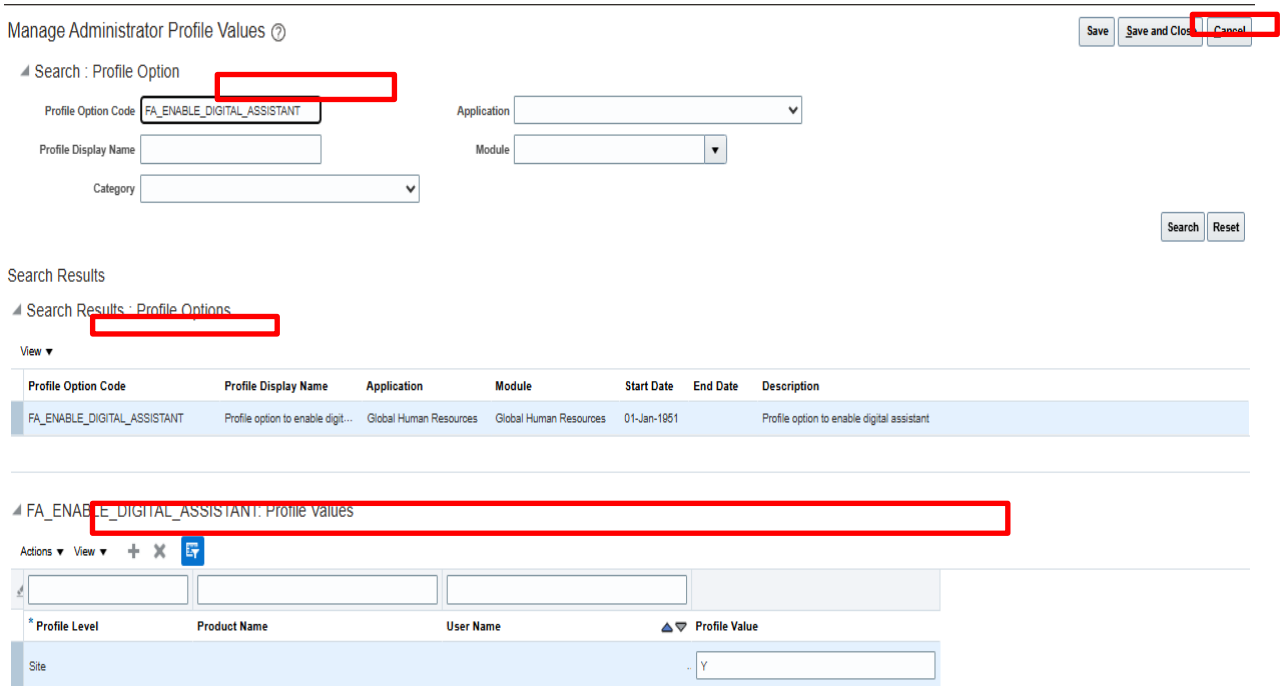
- After you configured the Oracle Web channel, set up the digital assistant to work with the Oracle Web channel and add the digital assistant in Oracle Fusion Cloud Applications. You can also enable attachment sharing for your users. After you enable it, users can share attachments, such as images, while interacting with the digital assistants.
- Enable the digital assistant for Fusion Applications.

    a. Sign in to Fusion Applications.

Navigation: FSM→Set & Maintenance→Global Search→ Manage Administrator Profile Values

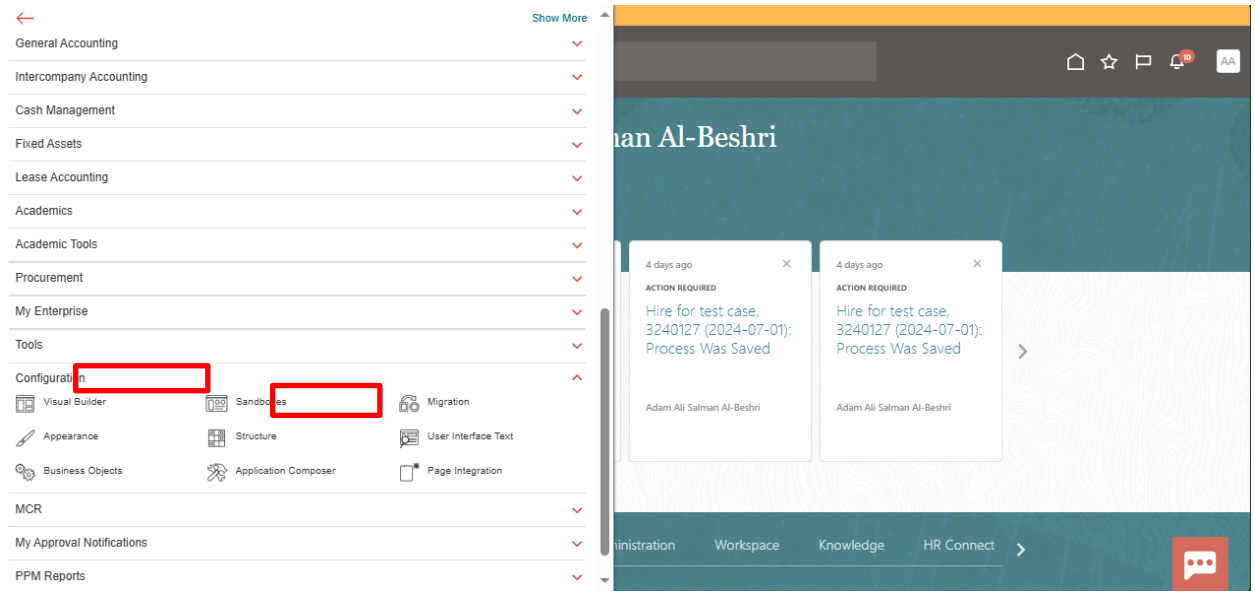


b. On the Manage Administrator Profile Values page, search for and select the profile option - Profile option to enable digital assistant (**FA_ENABLE_DIGITAL_ASSISTANT**).
c. From the Profile Level list, select **Site**.
d. In the Profile Value field, enter **Y**.
e. Click **Save and Close**.

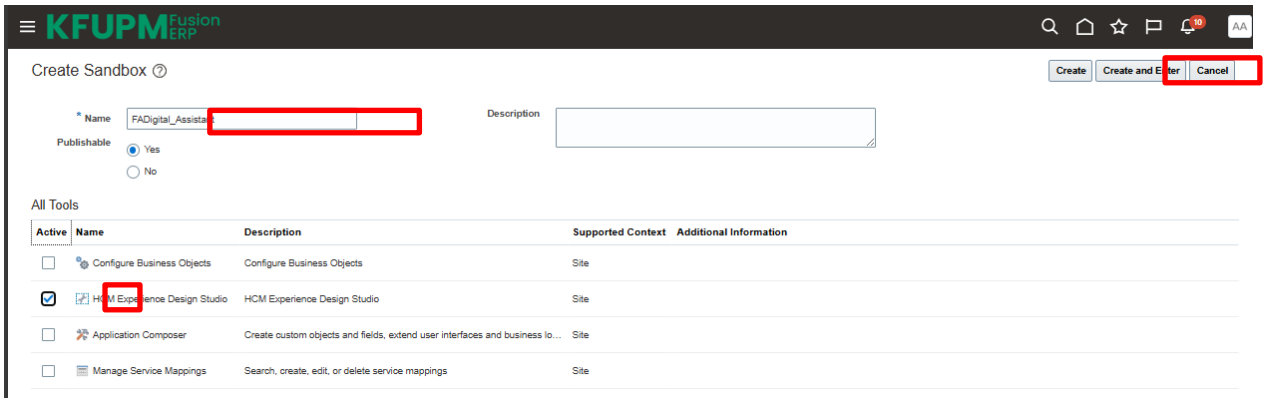


- **Add the digital assistant to Fusion Applications.**
- **Activate a sandbox that has the HCM Experience Design Studio tool in it.**

Navigation: Navigator→Configuration→Sandboxes→Create Sandbox

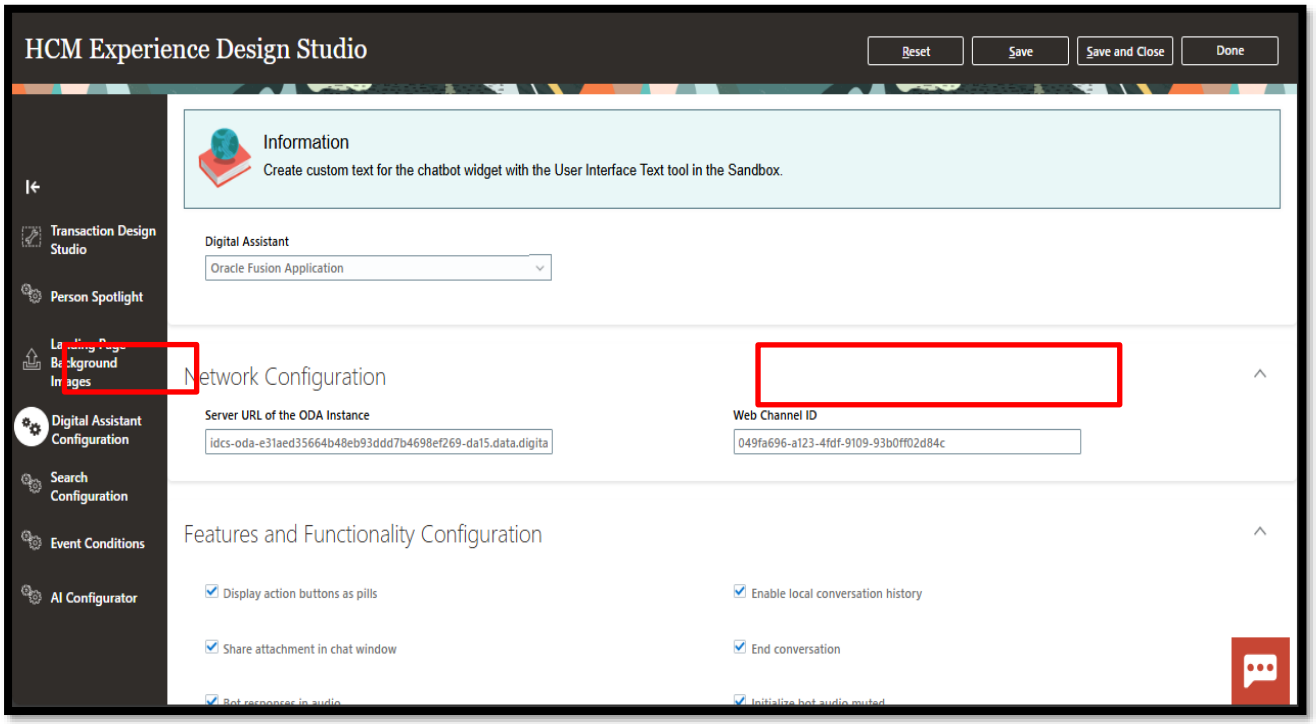


- **Enter name for sandbox, Select tool HCM experience design studio, click "Create and Enter".**

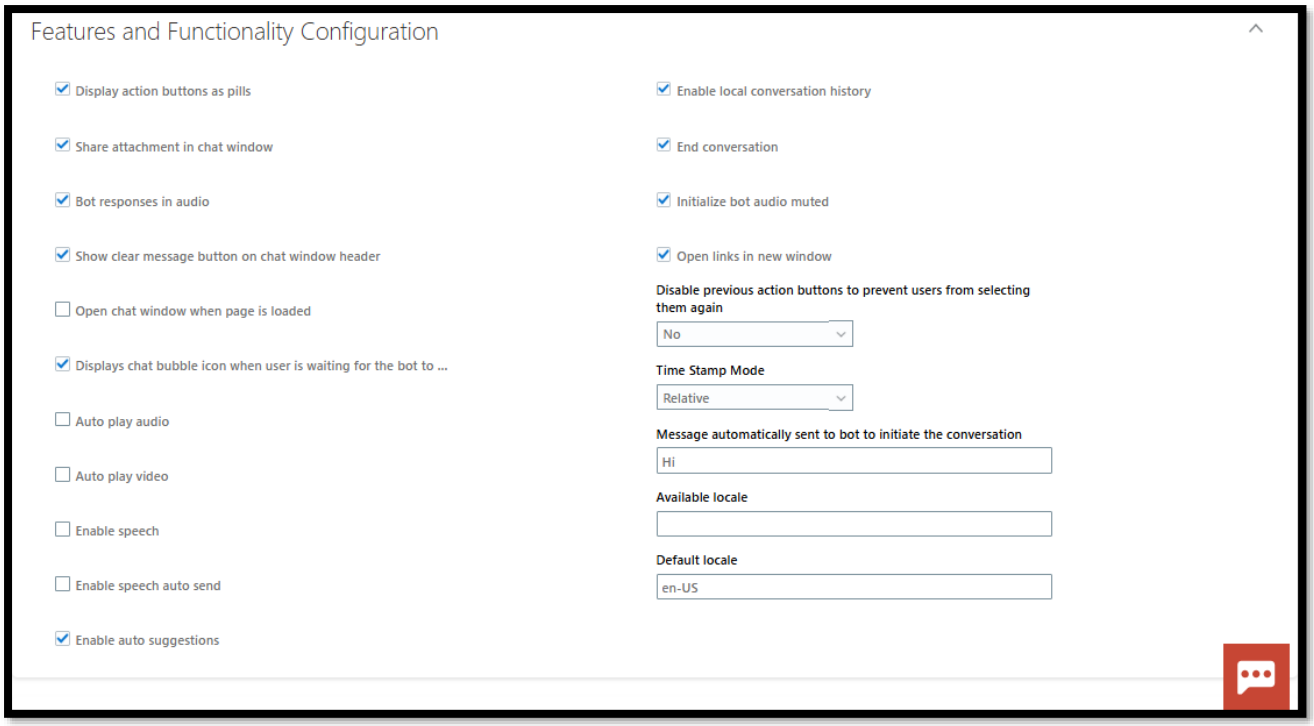


- On the Tools menu, select **HCM Experience Design Studio.**

- On the **HCM Experience Design Studio** page, click **Digital Assistant Configuration** tab.

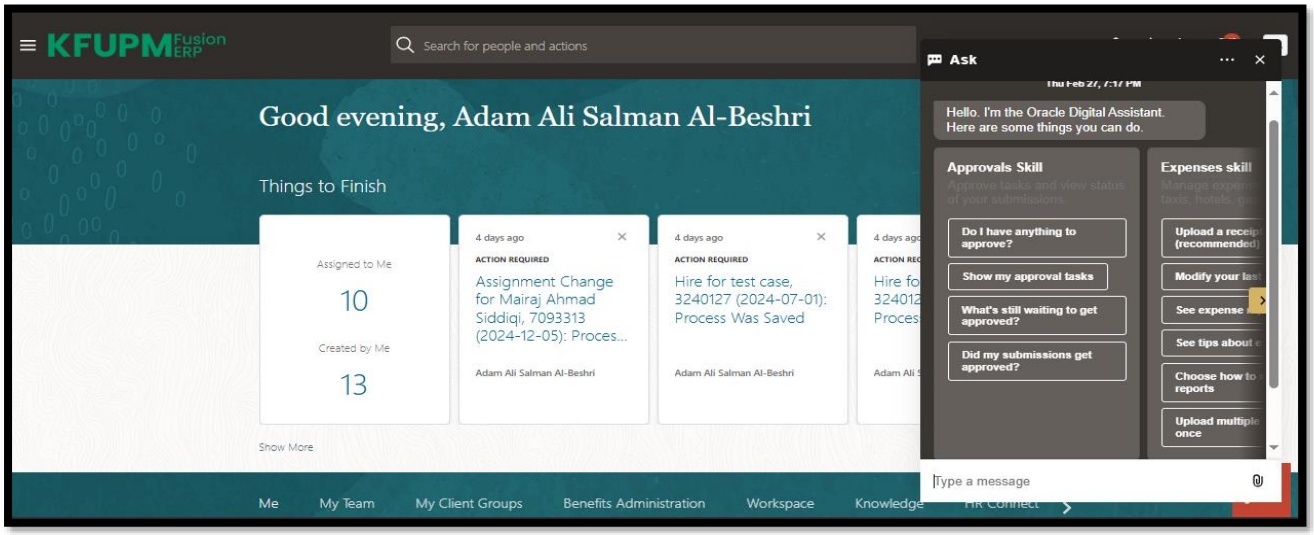


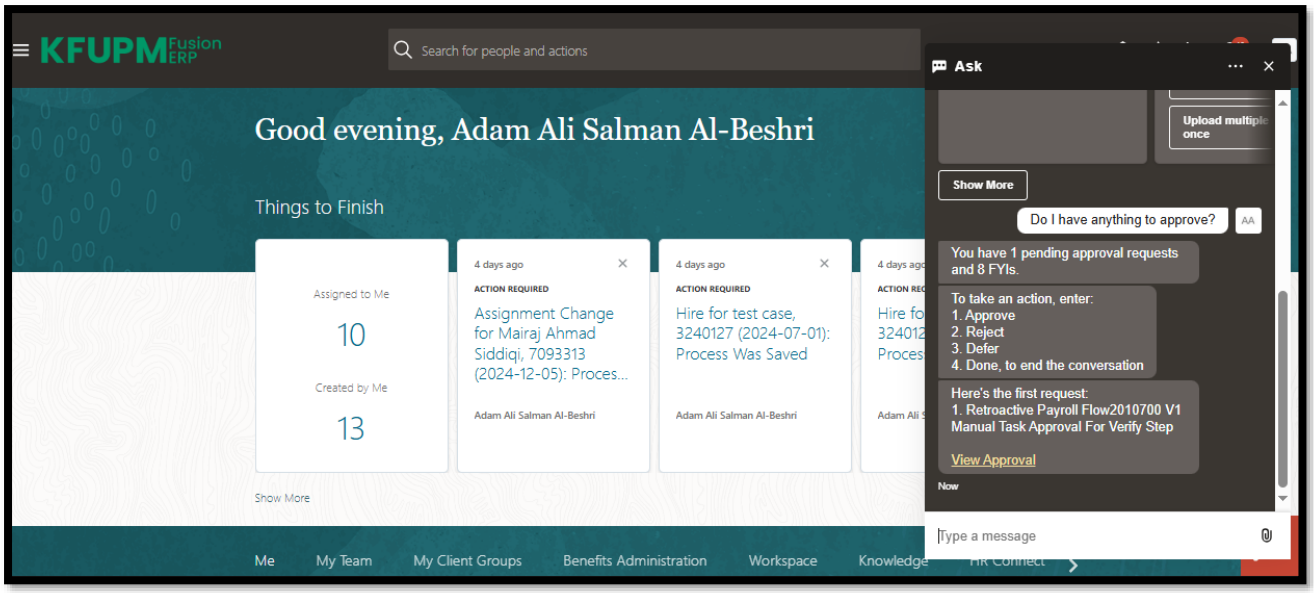
- In the **Web Channel ID** field, enter the web channel ID, which you got when you configured Oracle Web channel for your digital assistant.
- In the Features and Functionality Configuration section, specify other web channel preferences, such as enable attachment sharing.



- Click **Save and Close.**

5.Testing Digital Assistant:





- Test the configuration in the chat window and publish your sandbox.

Note :

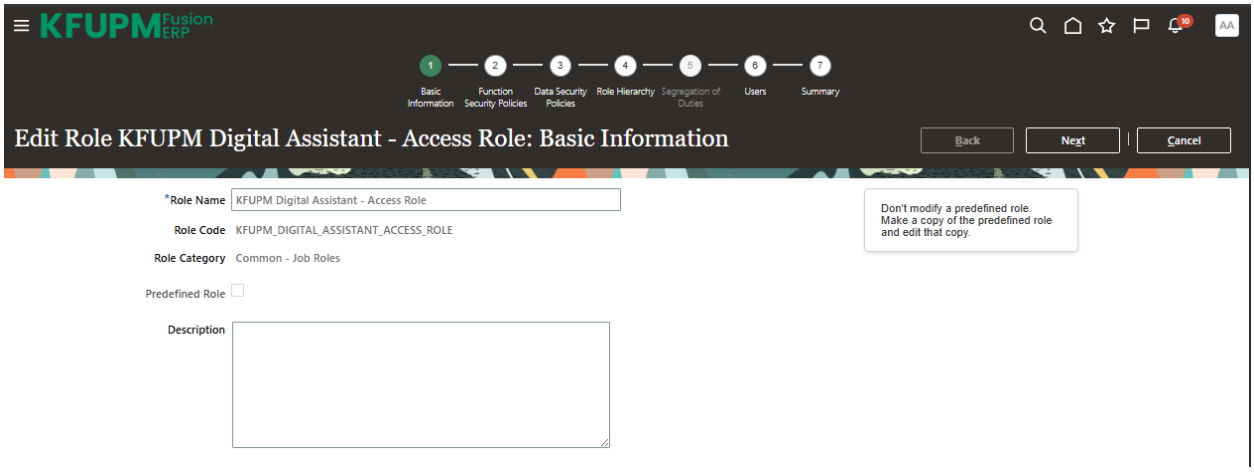
- To access the digital assistant, the final step involves obtaining the "**ACCESS DIGITAL ASSISTANT**" privilege.Assign this privilege to role

Privilege name: Access Digital Assistant
Code: HRC_ACCESS_DIGITAL_ASSISTANT_PRIV

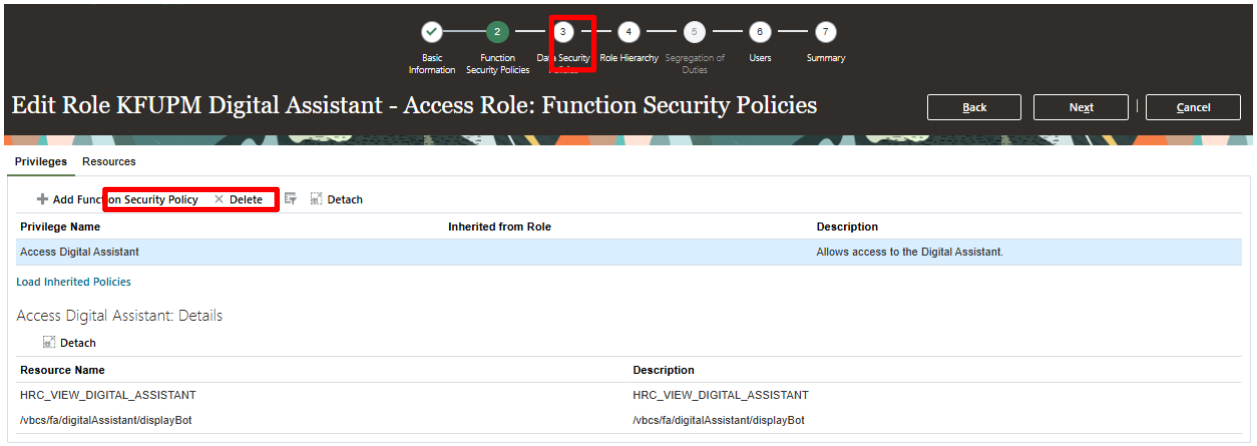
Creating a custom role for the digital assistant.

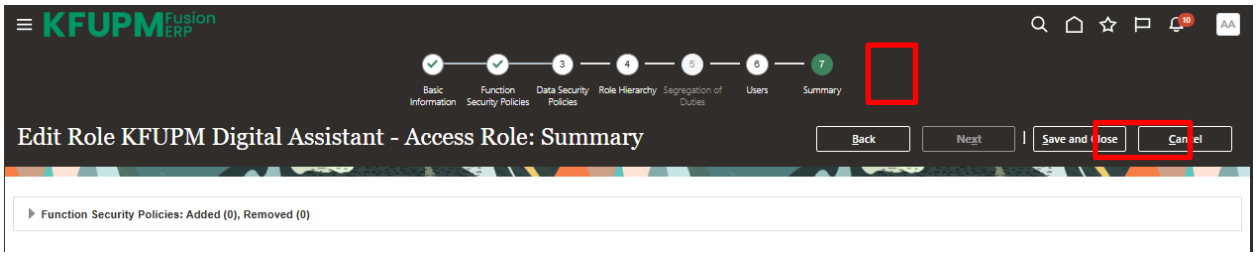
Navigation: Tools — Security Console — Roles — Create Role

a. Enter **Role Name**, **Role Code**, and choose Role Category: **Common - Job Roles**. Click **Next**.



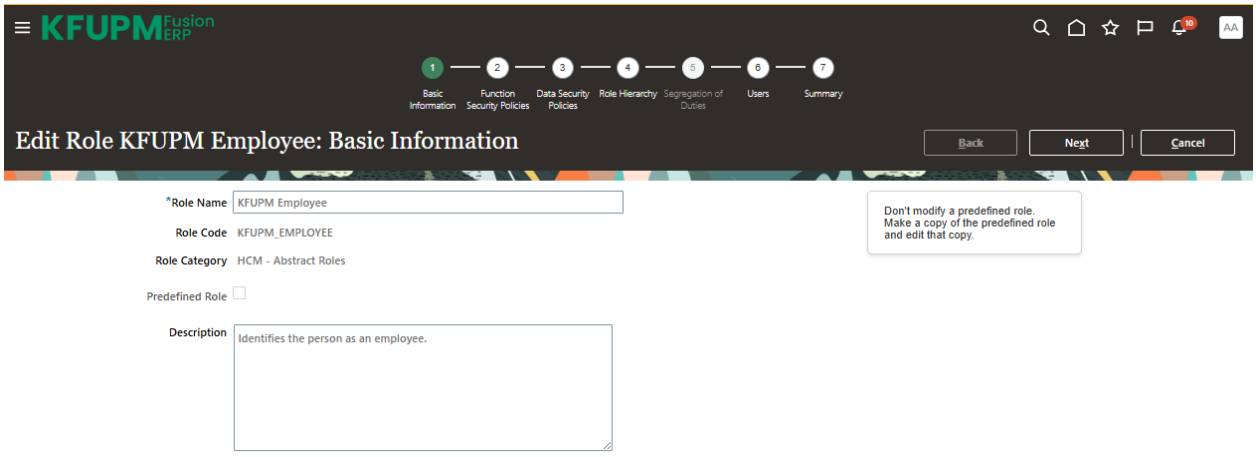
b. Add function privilege **Access Digital Assistant**, then click **Summary** and **Save and Close.**



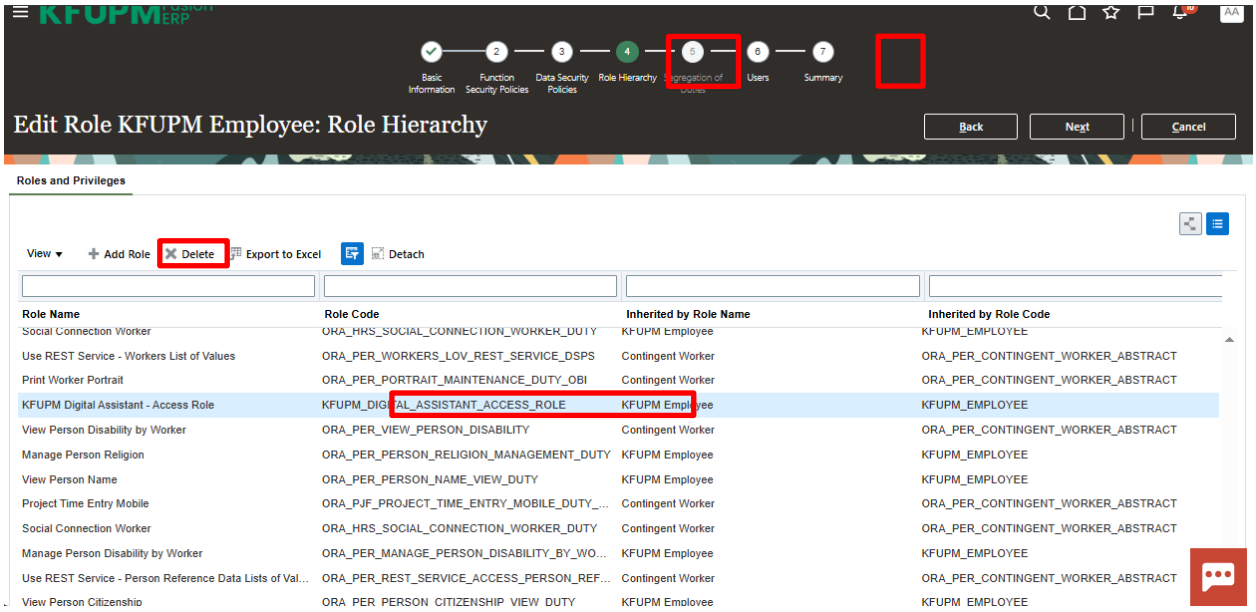


- Now, we can assign this role to other roles, such as the employee role or the line manager role.

Navigation: Tools — Security Console — Roles — KFUPM Employee



- Click **Add Role**. Choose the earlier created **Common - Job Roles for Access**.



- Click **Summary** then **Save and Close**.

- **INTEGRATING ODA WITH – WHATSAPP**

Overview

This document provides a step-by-step guide to integrating **Oracle Digital Assistant (ODA)** with **WhatsApp** using **Twilio**. By following this integration, the chatbot built in ODA can seamlessly send and receive messages over WhatsApp, enabling interactive conversational experiences.

TWILIO: Twilio is a cloud-based customer engagement platform that offers APIs to programmatically integrate communication features—including voice, messaging, video, email, and authentication—into applications. This eliminates the need to build communication infrastructure from scratch.

Architecture Overview

The integration involves the following interaction flow:

1. The user sends a message via WhatsApp.
2. Twilio receives the incoming WhatsApp message.
3. Twilio forwards the message to ODA via a configured webhook URL.
4. ODA processes the message through its dialog flows and bot logic.
5. ODA generates a response and sends it back via the webhook to Twilio.
6. Twilio delivers the response to the user on WhatsApp.

Prerequisites

Before initiating the integration, ensure the following requirements are met:

- Access to **Oracle Digital Assistant** with admin privileges.
- A **Twilio account** that supports WhatsApp (sandbox or live number).
- A valid domain with **HTTPS/SSL** configured for the ODA webhook listener.
- Basic knowledge of REST APIs and webhook concepts.

Key Terminology

Term	Definition
ODA	Oracle Digital Assistant
Twilio	Cloud communications platform for SMS, WhatsApp, etc.
Sandbox	Twilio's testing environment for WhatsApp integration before going live
Webhook	URL endpoint that receives incoming messages or events
Sender	WhatsApp number used to send messages via Twilio
Recipient / Dummy Recipient	WhatsApp number used for testing to receive messages
QR / Join Code	Code or QR mechanism used by a user to opt into the sandbox
Channel	Communication channel in ODA, in this case, Twilio WhatsApp via webhook

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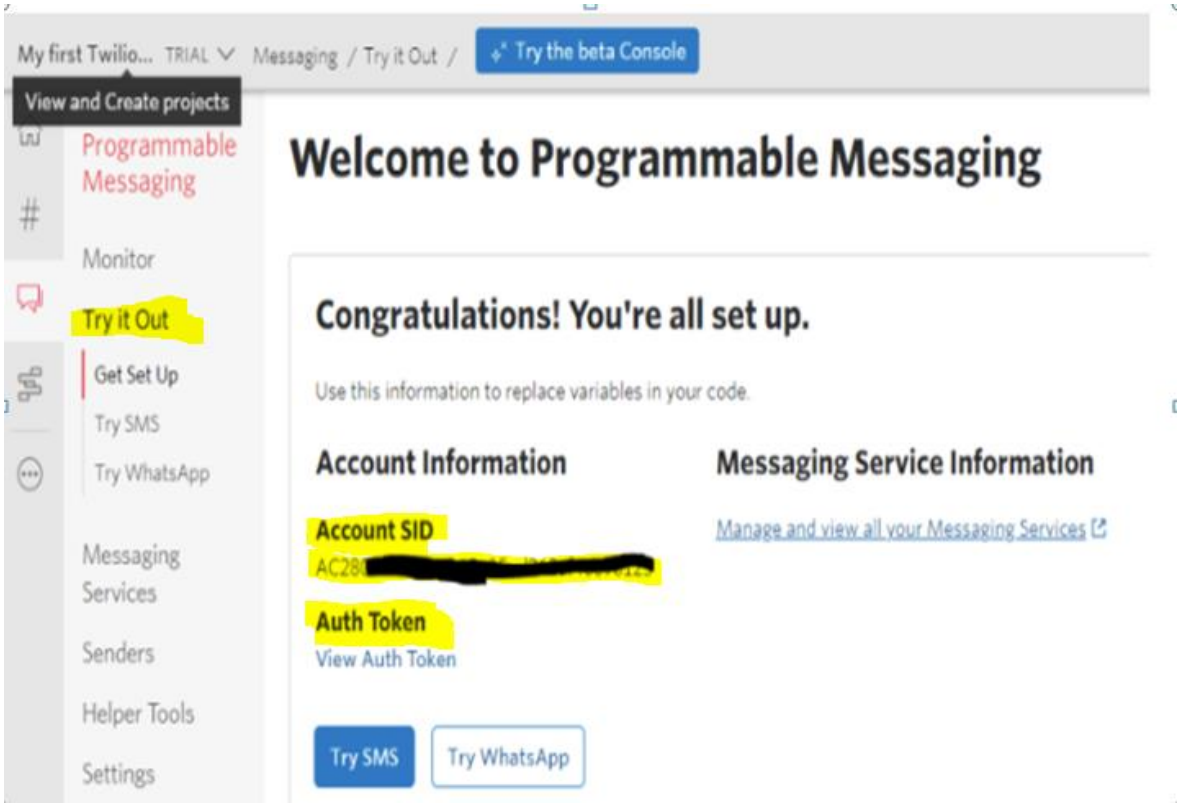
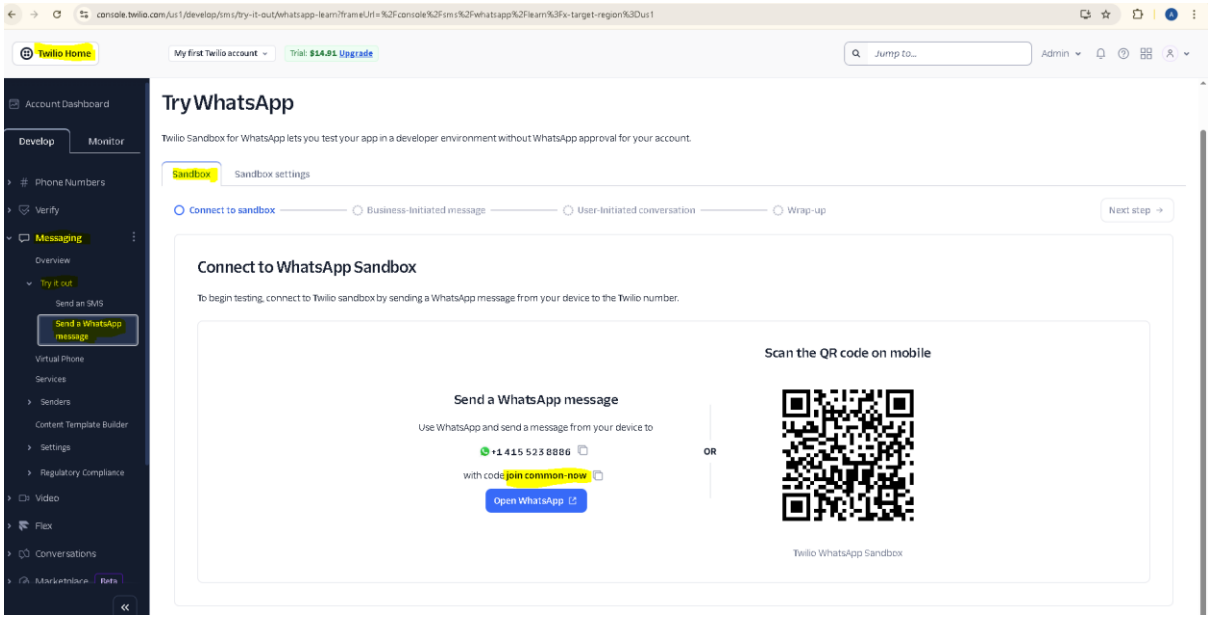
Step-by-Step Integration Process

1. Setting up Twilio Sandbox for WhatsApp

1. Log in to your **Twilio Console**.
2. Navigate to **Messaging → Try it Out → WhatsApp Sandbox**.
3. Set up your sandbox environment. Twilio provides:
   - Sandbox (sender) WhatsApp number (dummy number)
   - Join code or keyword (e.g., `join-common-now`) and QR code for recipient opt-in
4. Note the following details:
   - Sandbox sender number
   - Join code / keyword
   - Recipient dummy number for testing





2. Configuring WhatsApp Sender & Recipient

1. From the recipient number, send a WhatsApp message to the sandbox sender number using the join code (e.g., `join-common-now`). This opt-in step is required due to WhatsApp policies.
2. Once opted in, Twilio allows message exchange between the sandbox sender and the recipient.

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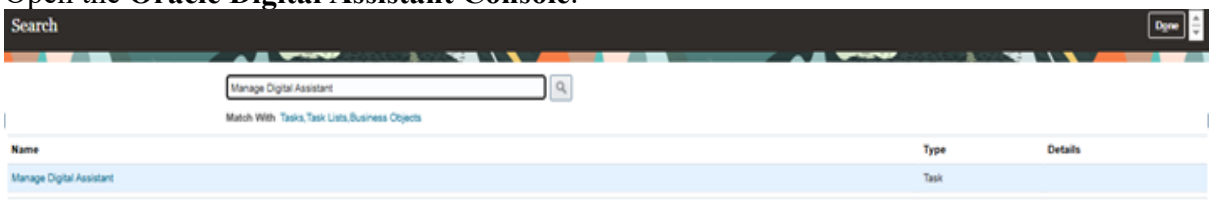
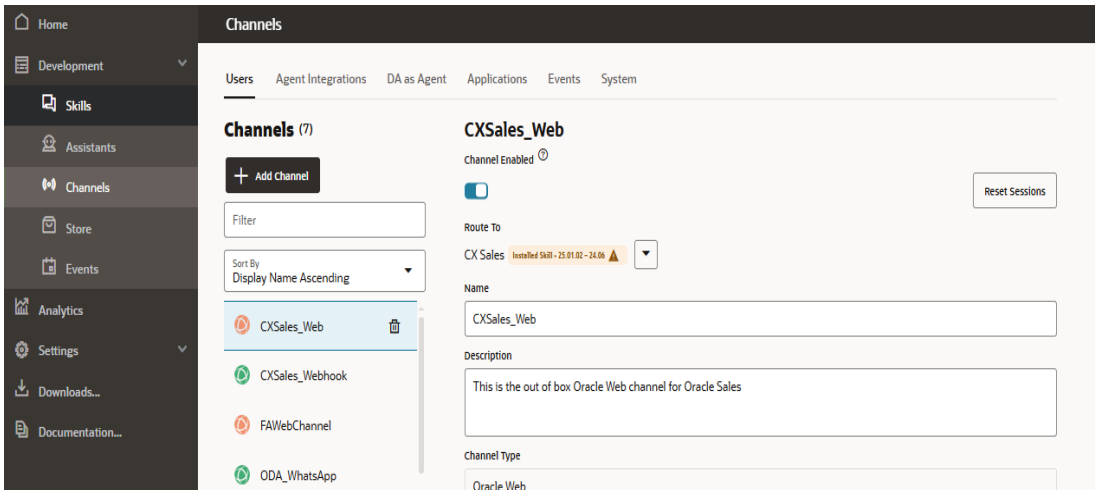
3. Creating & Configuring a Webhook Channel in ODA

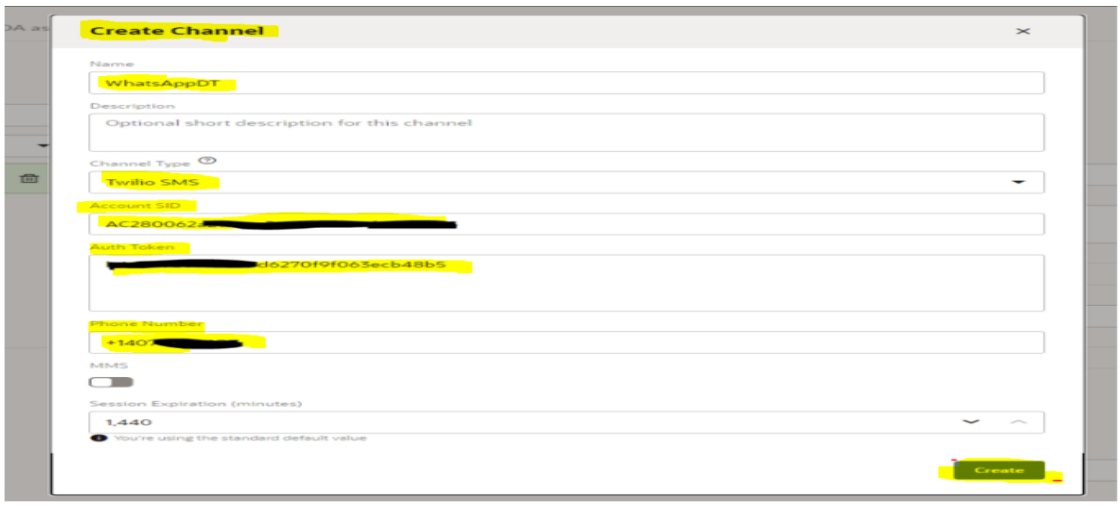
1. Open the **Oracle Digital Assistant Console**.



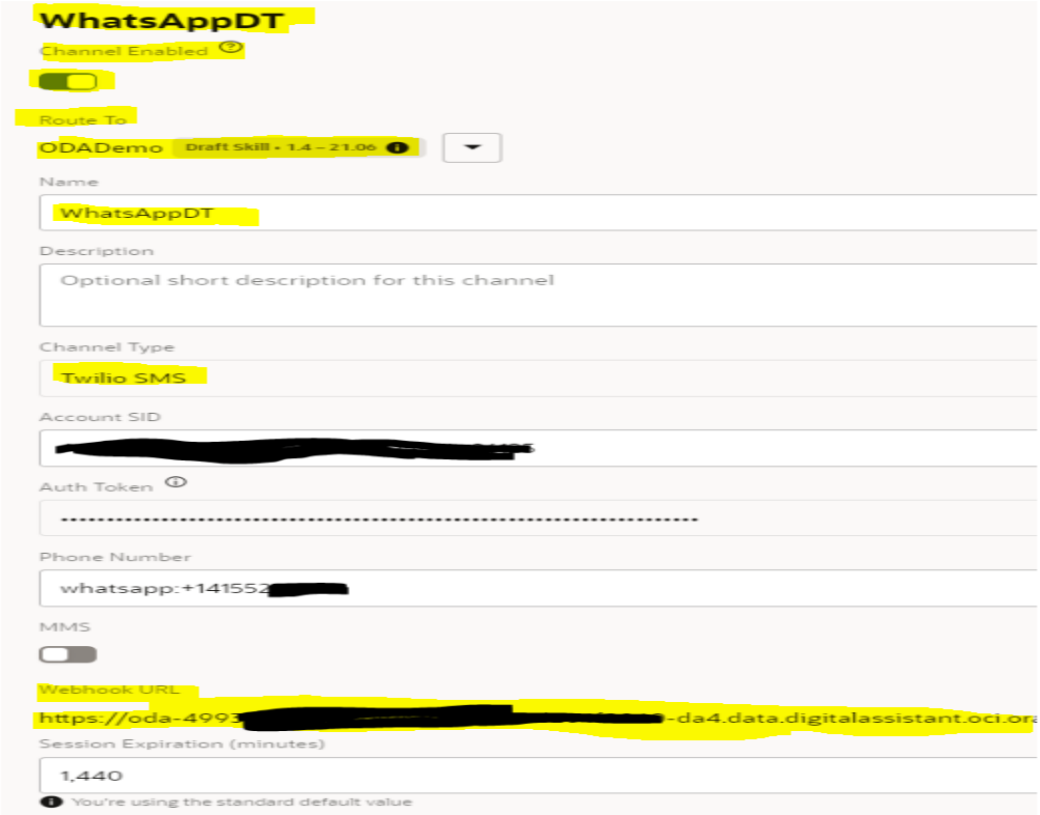
2.



3.
4. Navigate to **Channels → Add Channel** and select **Twilio WhatsApp** (or Custom Webhook depending on ODA version).
5. Enter the Twilio **Account SID** and **Auth Token**.
6. Enter the Twilio sandbox sender number.

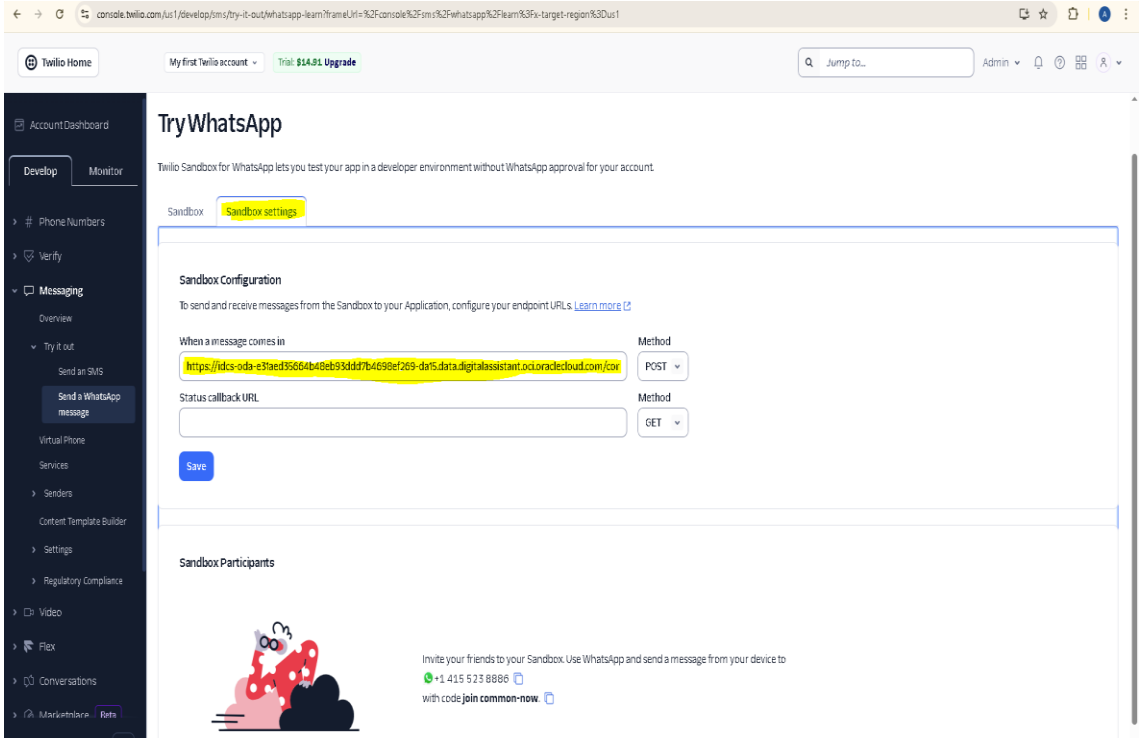
7. 

8. In the **Route To** property, select the ODA skill to which incoming messages should be routed.
9. Copy the generated **Webhook URL**. This URL will receive all incoming messages from Twilio.



4. Configuring the Webhook in Twilio

1. Navigate to **Twilio Console → Messaging → WhatsApp Sandbox → Sandbox Settings**.
2. In the field **"When a Message Comes In"**, paste the ODA webhook URL.
3. Save the settings.
4. Any message sent to the sandbox number will now be forwarded to ODA.



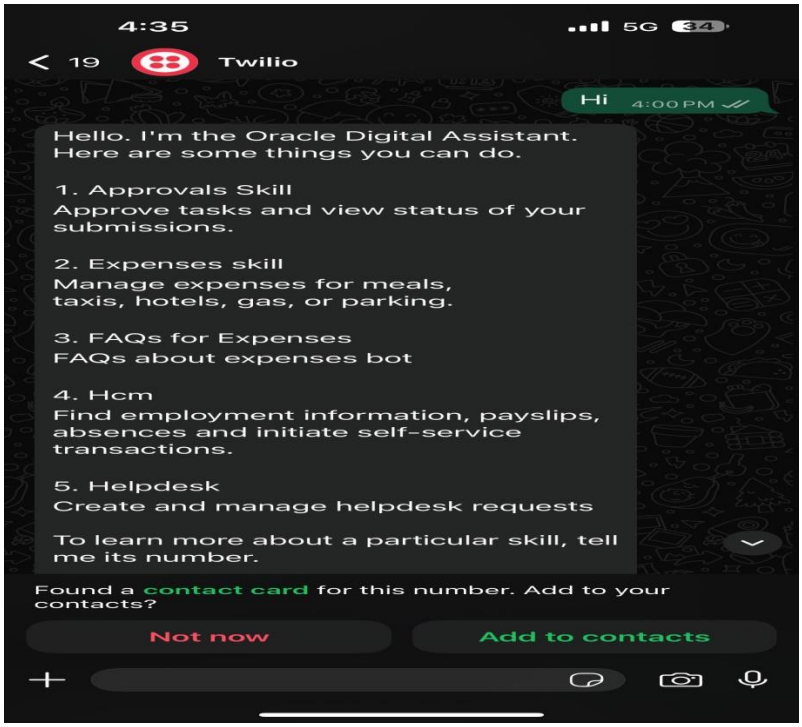
ODA Web hook URL: https://idcs-oda-e31aed35664b48eb93ddd7b4698ef269-da15.data.digitalassistant.oci.oraclecloud.com/connectors/v2/listeners/twilio/channels/e1748518-918e-4a24-b721-ca4b43444eb0

5. Testing the Integration

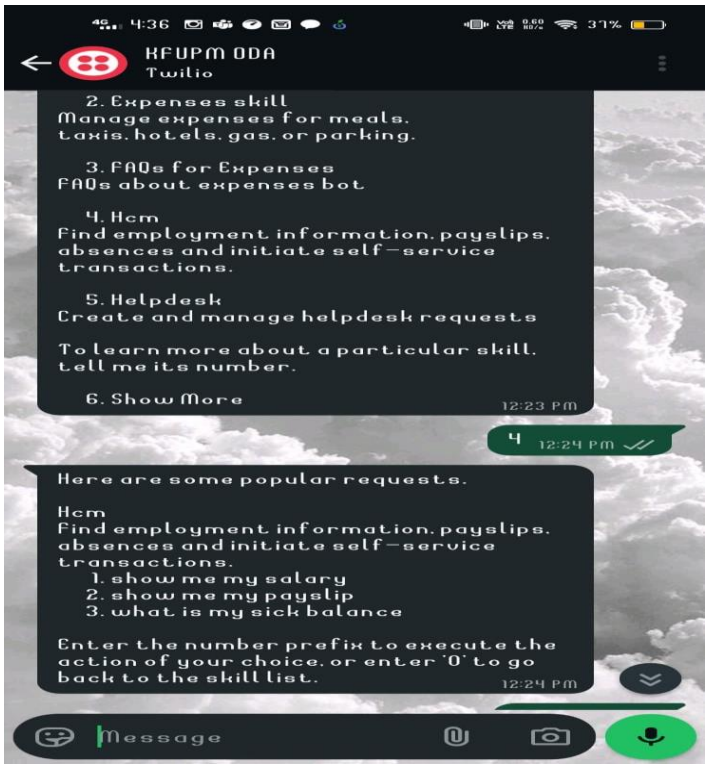
1. From the recipient (dummy) WhatsApp number, send a message to the Twilio sandbox number (e.g., "Hi").
2. Verify in Twilio logs that the message was forwarded to ODA.
3. In ODA, confirm that the message is received and processed according to the bot's dialog flow.
4. Verify that ODA responses are sent back via Twilio and delivered to the WhatsApp user.
5. Test additional scenarios such as message templates, errors, and media handling.

- We will send a 'Hi' message by scanning the QR code shown below. By providing the code received from Twilio, once it is confirmed, we will receive a confirmation message, as shown in the image below.

- Once the code is confirmed, we can access ODA by sending an initial dummy message. ODA will then respond according to the configuration defined in the skill within the ODA console. The responses are based on the skill to which the current channel is routed, as described in the steps above.



The image below illustrates that we are receiving responses from the chatbot (ODA) in WhatsApp.



Example Integration Flow

1. **Alice** sends `join-common-now` from her WhatsApp to the Twilio sandbox number to opt in.
2. She sends `Hi, what can you do?.`
3. Twilio forwards the message to ODA via the webhook.
4. ODA recognizes the greeting intent and responds: `Hello! I can assist you with X, Y, and Z.`
5. The response is sent back to Twilio via the webhook and delivered to Alice's WhatsApp.
6. Any buttons, attachments, or media from Alice are also forwarded to ODA following the same process.

Note :

- Twilio is a third-party service used to facilitate WhatsApp integration.
- ODA webhook URL must be added to the Twilio sandbox settings for message forwarding.
- The sender dummy number and recipient dummy number (or QR code) are used for testing purposes.